

ASSESSMENT TOOL - 6

1) SCHOOL MANAGEMENT SYSTEM - STUDENTS WITH CLASS TEACHER AND GRADE

```
mysql> SELECT s.Name, c.ClassTeacher, s.Grade
-> FROM Students s
-> JOIN Classes c ON s.ClassID = c.ClassID;
```

Name	ClassTeacher	Grade
Asha	Mrs. Sharma	A
Ravi	Mr. Kumar	B
Priya	Mrs. Sharma	A
Karan	Mr. Singh	B

4 rows in set (0.00 sec)

2) HOSPITAL - DOCTORS WITH APPOINTMENTS THIS MONTH

```
mysql> SELECT d.DoctorName, COUNT(a.AppointmentID) AS ApptCount
-> FROM Doctors d
-> JOIN Appointments a ON d.DoctorID = a.DoctorID
-> WHERE MONTH(a.AppointmentDate) = MONTH(CURDATE())
-> GROUP BY d.DoctorID, d.DoctorName
-> HAVING COUNT(a.AppointmentID) > 10;
```

DoctorName	ApptCount
Dr. Mehta	12
Dr. Sharma	14

2 rows in set (0.00 sec)

3) E- COMMERCE - PRODUCTS NEVER ORDERED (SUB QUERY)

```
mysql> SELECT ProductName
-> FROM Products
-> WHERE ProductID NOT IN
-> (SELECT DISTINCT ProductID FROM Orders);
+-----+
| ProductName |
+-----+
| Wireless Mouse |
| Smartwatch |
| Desk Lamp |
+-----+
3 rows in set (0.00 sec)
```

4) UNIVERSITY - STUDENTS ENROLLED IN BOTH "DBMS" AND "OS"

```
mysql> SELECT s.Name
-> FROM Students s
-> WHERE s.SID IN
- (SELECT e.SID FROM Enrolled e
  WHERE e.CID = (SELECT CID FROM Courses WHERE CName = 'DBMS')
  INTERSECT
  SELECT e.SID FROM Enrolled e
  WHERE e.CID = (SELECT CID FROM Courses WHERE CName = 'OS'));
+-----+
| Name |
+-----+
| Rahul |
+-----+
1 row in set (0.00 sec)
```

5) PAYROLL - VIEW AND UPDATE SALARY (10% INCREASE)

```
mysql> CREATE VIEW LowSalaryEmp AS
  -> SELECT EmpID, EmpName, Salary, DeptID
  -> FROM Employee
  -> WHERE Salary < (SELECT AVG(Salary) FROM Employee GROUP BY DeptID);
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM LowSalaryEmp;
+-----+-----+-----+-----+
| EmpID | EmpName | Salary | DeptID |
+-----+-----+-----+-----+
| 101   | Karan   | 40000  | 2      |
| 104   | Neha    | 45000  | 1      |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> UPDATE Employee
  -> SET Salary = Salary * 1.10
  -> WHERE EmpID IN (SELECT EmpID FROM LowSalaryEmp);
Query OK, 2 rows affected (0.01 sec)
```

6) BANKING - CUSTOMER, ACCOUNT, TRANSACTION, BRANCH DETAILS

```
mysql> SELECT c.CustomerName, a.AccountNo, t.Amount, t.TransactionDate,
  -> b.BranchName
  -> FROM Customers c
  -> JOIN Accounts a ON c.CustomerID = a.CustomerID
  -> JOIN Transactions t ON a.AccountID = t.AccountID
  -> JOIN Branches b ON a.BranchID = b.BranchID;
+-----+-----+-----+-----+-----+
| CustomerName | AccountNo | Amount | TransactionDate | BranchName |
+-----+-----+-----+-----+-----+
| Arun         | 1001      | 5000   | 2024-05-10      | Main Branch |
| Priya        | 1002      | 7000   | 2024-05-11      | City Branch |
| Rahul        | 1003      | 2000   | 2024-05-12      | Main Branch |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

7) RETAIL STORE - DAILY SALES SUMMARY (GROUP BY, HAVING)

```
mysql> SELECT Category, SUM(Amount) AS TotalSales
-> FROM Sales
-> GROUP BY Category
-> HAVING SUM(Amount) > 10000;
```

Category	TotalSales
Electronics	12500
Clothing	12000
Home	15000

3 rows in set (0.00 sec)

8) LIBRARY - MEMBERS BORROWED BOOKS FROM ALL GENRES(TUPLE RELATIONAL CALCULUS)

```
mysql> SELECT M.Name
FROM Members M
WHERE NOT EXISTS (
  SELECT G.Genre
  FROM Genres G
  WHERE NOT EXISTS (
    SELECT * FROM Borrowed B
    WHERE B.MemberID = M.MemberID
    AND B.Genre = G.Genre
  );
```

Name
Suresh

1 row in set (0.00 sec)

9) INVENTORY - CONSTRAINTS AND ENFORCEMENT

```
mysql> CREATE TABLE Product
-> (
->   ProductID INT PRIMARY KEY,
->   ProductName VARCHAR(50) NOT NULL,
->   CategoryID INT,
->   CONSTRAINT fk_category
->   FOREIGN KEY (CategoryID) REFERENCES Category(CategoryID),
->   CONSTRAINT chk_price
->   CHECK (Price > 0)
-> );
Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO Product VALUES (1,'Laptop',1,50000);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Product VALUES (2,'Cheap Pen', 1, -10);
ERROR 97 (23000): Check constraint 'chk_price' is violated.
```

10) HR SYSTEM - MATERIALISED VIEWS

```
mysql> CREATE MATERIALIZED VIEW MonthlyAttendance AS
-> SELECT e.EmpID, e.EmpName, COUNT(a.AttendanceDate) AS DaysPresent
-> FROM Employees e
-> JOIN Attendance a ON e.EmpID = a.EmpID
-> WHERE MONTH(a.AttendanceDate) = MONTH(CURDATE())
-> GROUP BY e.EmpID, e.EmpName;
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE MATERIALIZED VIEW MonthlyPerformance AS
-> SELECT e.EmpID, e.EmpName, AVG(p.Rating) AS AvgRating
-> FROM Employees e
-> JOIN Performance p ON e.EmpID = p.EmpID
-> WHERE MONTH(p.ReviewDate) = MONTH(CURDATE())
-> GROUP BY e.EmpID, e.EmpName;
Query OK, 0 rows affected (0.05 sec)
```